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Original article

CYBER-ATTACK VIA SOCIAL ENGINEERING IN ISRAEL: A CASE STUDY OF THE HAMSAUPDATE MALWARE CAMPAIGN

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Abstract: *This case study examines the HamsaUpdate malware campaign, a deliberate cyber assault on Israeli civilians amid the 2023-2024 Israel-Hamas conflict. The operation disseminated a deceptive emergency alert application via WhatsApp, aiming to install spyware on users' mobile devices under the pretence of providing critical information. This study analyses the campaign's operational mechanisms, psychological effects, and institutional responses through qualitative analysis of technical reports, cybersecurity bulletins, and academic literature. The findings, contextualised within hybrid warfare and social engineering theories, demonstrate how adversaries leveraged public trust in digital emergency tools to fulfil surveillance and destabilisation objectives. The assault marked a tactical advancement in cyberwarfare, shifting from rudimentary propaganda and defacement to more clandestine and invasive espionage. Malicious APK files facilitated real-time data exfiltration, including camera, microphone, location, and messaging content. The use of social media and linguistic modification significantly enhanced the campaign's reach and effectiveness. The psychological effects, characterised by confusion, fear, and diminished trust in official communication channels, intensified societal vulnerability during physical conflict. This study contributes to cybersecurity readiness by highlighting the convergence of technical exploitation and psychological manipulation in modern conflicts. Key policy implications include the immediate adoption of zero-trust architecture, real-time threat intelligence dissemination, enhanced digital literacy, and simulation-based training to prepare civilians for socially engineered threats. The HamsaUpdate campaign exemplifies the evolution of citizen-focused cybersecurity strategies within hybrid conflict contexts.*

Keywords: Hybrid warfare, Social Engineering, Mobile Malware, Digital Trust, Cyber Assault, HamsaUpdate

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1. Introduction

Cyberattacks have become strong instruments of geopolitical influence, psychological disruption, and strategic conflict in the digital age. They have gone beyond basic data breaches and have become far more sophisticated. Hybrid methods are becoming increasingly prevalent in modern warfare. These strategies entail the integration of traditional military power with cyber operations that attack both infrastructure and society (Kostyuk & Zhukov, 2019). Social engineering assaults are among the most successful methods within this paradigm. These attacks take advantage of psychological vulnerabilities, emotional stress, and cognitive biases in order to trick individuals into divulging information or carrying out acts that are damaging (Taddeo & Floridi, 2018).

With the increasing digital interconnection of societies, cyberattacks are no longer limited to affecting only state systems or organisations; rather, they are increasingly able to directly impact the lives of normal citizens (Carvalho et al., 2023). More than sixty per cent of the most recent cyber incidents that occurred in the European Union involved the manipulation of human behaviour (ENISA, 2023). This digital

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dependence is exploited by enemies during times of war, when public fear is at its highest, in order to propagate disinformation and spyware, which can destabilise both trust and response capability (Linkov & Kott, 2019).

One particularly dramatic example of the convergence of psychological warfare and cyber espionage was offered by the armed conflict between Israel and Hamas in the years 2023 and 2024. The HamsaUpdate effort, which was one of the most prominent cyber operations of the conflict, involved the distribution of a false emergency alert application through WhatsApp. The programme was disguised as a civil defence tool (Intezer, 2023). In reality, it was spyware that accessed the microphone, camera, location, and messages of users (Check Point Research, 2024; Intezer, 2023). As rocket assaults were being carried out, the operation targeted Israeli citizens, taking advantage of their panic and sense of urgency to increase download rates. This strategy is in line with what is referred to as "cognitive warfare", which is an intentional endeavour to influence the feelings and actions of civilians via the use of technological deception (Henschke, 2024).

In spite of the rising body of research on cyberwarfare, a very small number of studies have investigated the ways in which malware campaigns influence people directly during times of active armed conflict. While the majority of the research that has been conducted on this topic has concentrated on infrastructure or national security (Kostyuk & Zhukov, 2019), the psychological impact that these digital attacks have had on civilians has not been thoroughly investigated (Bean & Grevstad, 2023).

The purpose of this study is to bridge that gap by posing the following questions: how did the HamsaUpdate malware campaign use social engineering techniques during the Israel-Hamas conflict to weaken civilian digital trust, and what are the consequences for cybersecurity preparation in hybrid warfare contexts? The research employs a qualitative document analysis of technical cybersecurity reports, governmental warnings, and peer-reviewed academic literature. This analysis is guided by theoretical frameworks of hybrid warfare and social engineering. The study explores three important aspects: (1) the operational design of the virus; (2) the psychological and behavioural consequences for civilians; and (3) the gaps in institutional responses and readiness. Through the examination of HamsaUpdate as a case of socially engineered cyberwarfare, this research contributes to the existing body of literature on the resilience of civilian cybersecurity.

2. Theoretical Background

This study is based on three related areas of theory: hybrid warfare, social engineering, and digital trust and civilian resilience. Together, these theories explain how modern cyber campaigns combine technology with psychological and behavioural manipulation to erode individuals' sense of safety and destabilise society.

Hybrid warfare theory asserts that contemporary conflicts increasingly conflate military, cyber, economic, and informational domains (Fridman et al., 2019). Instead of solely engaging in conventional kinetic warfare, adversaries utilise a comprehensive approach encompassing cyberattacks, disinformation, psychological operations, and economic disruption (Libicki, 2020). This multidimensionality is particularly effective in asymmetric conflicts, as state and non-state actors exploit weaknesses in civilian digital infrastructure as a force multiplier (Linkov & Kott, 2019). Within this perspective, malware transcends its role as a mere technical tool, evolving into an instrument of geopolitical influence and psychological destabilisation (Linkov & Kott, 2019; Manolea, 2021).

The theory of social engineering enhances the hybrid warfare paradigm by emphasising the manipulation of human behaviour through deception and emotional exploitation (Taddeo & Floridi, 2018). In cybersecurity, social engineering refers to a category of attacks that exploit the "human factor" rather than technological flaws (Hadnagy & Schulman, 2021). These strategies often employ impersonation, urgency, fear, or authoritative signals to compel users to click harmful links, install malware, or disclose confidential information. In combat scenarios, such strategies become particularly potent, as fear and confusion increase susceptibility to manipulation (Hadnagy & Schulman, 2021). The HamsaUpdate

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campaign, masquerading as a civil defence application, exemplifies the intersection of emotional manipulation and technological deception (Check Point Research, 2024; Intezer, 2023).

Digital trust and civilian resilience frameworks elucidate individual responses to cybersecurity threats, particularly in emergency contexts. Confidence in official digital services, such as emergency alert systems or governmental communication channels, is essential for effective crisis management (ENISA, 2023). When cyberattacks impersonate or exploit trusted systems, they can undermine public confidence, reduce system compliance, and intensify panic (Shandler & Gomez, 2023). Civilian resilience in cyberspace requires not only robust technical infrastructure but also adaptive behavioural responses, digital literacy, and coordinated institutional action (Linkov et al., 2013). Zero-trust architecture, which holds that no implicit trust should be granted to any system component or user, has emerged as both a policy and design philosophy aimed at countering manipulation-based assaults (CISA, 2022; ENISA, 2023).

This theoretical paradigm positions the HamsaUpdate malware campaign within a multifaceted conflict strategy that integrates technological infiltration, behavioural manipulation, and information distortion. By combining insights from these domains, the study offers a comprehensive analytical framework for understanding the growing risks cyber operations pose to civilian populations in hybrid warfare.

3. Methods

This case study employs a qualitative document analysis method to explore the HamsaUpdate malware campaign, which emerged during the Israel-Hamas conflict from 2023 to 2024. When it is not possible to reach affected individuals, access proprietary systems, or obtain private technical data, document analysis serves as a useful approach for investigating digital security challenges. This method allows for the systematic examination of various texts to identify patterns, draw conclusions, and understand events within their broader context (Bowen, 2009).

The data corpus comprised publicly available documents from 2023 to 2025. These included threat intelligence reports from cybersecurity companies such as Check Point Research and Intezer, government cybersecurity advisories, peer-reviewed academic papers on cyberwarfare and social engineering, and verified media reports. These sources were selected through purposive sampling, focusing on those that directly relate to the HamsaUpdate campaign or provide relevant theoretical and operational context. The criteria for source selection included credibility, recency, relevance to the research topic, and the depth of analysis provided. For instance, sources containing technical details about the HamsaUpdate virus (e.g., how it operates, how it exfiltrates data, and how it spreads via apps) were prioritised, as were academic papers examining the use of psychological manipulation in cyber campaigns during warfare.

The analytical methodology employed was thematic content analysis, involving the identification of recurring codes and themes across multiple documents (Braun & Clarke, 2006). This included both explicit content, such as specific malware characteristics, and implicit content, such as underlying behavioural strategies or implications for civilian trust. Data were coded in multiple cycles and organised into key thematic areas: operational mechanisms (e.g., dissemination of malicious APKs or use of WhatsApp), psychological and behavioural responses (e.g., panic or distrust in digital platforms), and institutional or policy-level actions (e.g., government alerts or cybersecurity education initiatives). To strengthen the credibility and validity of the findings, triangulation was applied across different source types, including technical reports, academic studies, and government publications.

To enhance the reliability of the research, clear and consistent procedures were followed. A log of data extraction was maintained, and source triangulation was used to avoid over-reliance on any single account. In addition, memoing techniques were employed to track analytical decisions, emergent insights, and potential researcher bias throughout the process. Nonetheless, several methodological limitations must be acknowledged. First, the study does not include the collection of primary data, such as interviews with affected civilians, cybersecurity professionals, or app platform moderators. This limits the ability to directly assess user experiences and decision-making processes. Second, certain technical aspects of the malware remain classified or proprietary, restricting the depth of forensic analysis. Third, cyber attribution remains

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a contentious and evolving field. As new evidence emerges, conclusions regarding the campaign's origin, objectives, or scope may be subject to revision. Despite these limitations, the use of a broad range of high-quality sources and a theoretically grounded analytical approach strengthens the credibility and depth of the findings. It also lays the groundwork for future research adopting more empirical and multi-method perspectives on cyber campaigns.

4. The Results and a Description of the Case

The HamsaUpdate malware campaign was a coordinated cyber operation conducted during the 2023-2024 conflict between Israel and Hamas. This study revealed key insights into the campaign's characteristics and technical deployment. Following the completion of thematic document analysis, several interrelated findings emerged. These are presented through a structured case description and thematic discussion. The case of HamsaUpdate offers a concrete example of hybrid warfare, characterised by the convergence of social engineering, mobile malware, and information manipulation aimed at undermining national resilience.

4.1 A Description of the Case: The Malware Campaign Based on HamsaUpdate

A covert cyber campaign targeting civilians was uncovered by Israeli cybersecurity authorities and international firms in the fourth quarter of 2023. This campaign, known as Operation HamsaUpdate, was executed via a fraudulent emergency application bearing the same name. WhatsApp served as the primary channel for malware dissemination, with messages circulated in both Hebrew and Arabic. The malware was disguised as an official civil defence alert app (Barnea, 2024). Its objective was to exploit the psychological stress experienced by civilians during rocket attacks by prompting them to download a tool that appeared legitimate and potentially life-saving (Check Point Research, 2024).

The malicious software harvested sensitive information from infected devices, including live microphone and camera feeds, geolocation data, and content from messaging applications. The campaign employed several high-level tactics, such as a rotating command-and-control infrastructure, encryption, and code obfuscation, which hindered early detection (Check Point Research, 2024; Intezer, 2023).

Attribution of the campaign was directed towards a group calling itself the Handala Hack Team, which surfaced during the 2023-2024 Israel-Hamas conflict. While the group portrays itself as a pro-Palestinian hacktivist collective, cybersecurity experts suspect it may be linked to a state actor or more organised entity. This assessment stems from the campaign's advanced technical capabilities, use of Hebrew-language content, and alignment with broader political objectives. Nevertheless, precise attribution remains difficult due to the inherent complexities of cyber forensics (Check Point Research, 2024; Intezer, 2023). The attackers appeared to be motivated not by financial gain, but by objectives related to surveillance and psychological destabilisation (Intezer, 2023).

Table 1 presents the main actors involved in the HamsaUpdate campaign, the methods they employed, and the resulting effects. This overview underscores the campaign's complexity and its socio-technical implications.

Table 1. Key Actors, Methods, and Effects of the HamsaUpdate Malware Campaign

Actors	Methods	Effects
Hamas-affiliated cyber units	-Social engineering via WhatsApp -Fake emergency alert app -Malicious APK files	-Data theft (GPS, messages, camera/mic) -Undermined trust in official -Apps psychological distress
Israeli National Cyber Directorate (INCD) & Computer Emergency Readiness Team-Israel (CERT-IL)	-Public alerts and advisories -Coordination with telecom firms -Publishing Indicators of Compromise (IoCs)	-Mitigated further infections -Launched digital hygiene -Campaigns -Raised awareness
Cybersecurity vendors (Check Point, Intezer)	-Malware analysis -Infrastructure tracking - Technical attribution reports	-Provided technical insights -Assisted in attribution to Hamas -linked groups

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Civilians	-Downloaded malware due to fear and misinformation	-Devices infected -Breach of privacy -Distrust in emergency tools
App marketplaces & platforms	-Removal of malicious links -Updates to security policies	-Reduced app spread -Improved future detection

4.2 Mechanisms of Operational Function: Techniques of Delivery and Technical Characteristics

To disseminate malicious APK files, the operation employed sophisticated distribution channels that bypassed official app stores. Specifically, it targeted peer-to-peer platforms and social media networks. References to "Red Alerts", military attacks, and civilian shelters were embedded in the messages, reinforcing their urgency. Users were misled into believing that the application was endorsed by government agencies, owing to fake branding that closely imitated the Israeli Home Front Command. These social engineering tactics enhanced the perceived legitimacy of the malware, resulting in a high click-through and installation rate (ENISA, 2023).

The malware demonstrated an elevated level of technical sophistication, including rootkit-level access, dynamic server rerouting, and real-time sensor activation. Such capabilities indicate a shift beyond mere data theft towards strategic surveillance and psychological influence (ENISA, 2023). The attackers' emphasis on reach and credibility was further evidenced by the use of culturally and linguistically tailored content (Check Point Research, 2024; Intezer, 2023).

4.3 Psychological and Behavioral Impact on Civilians

A significant psychological impact was exerted on the Israeli public as a result of the campaign, particularly in frontline cities located near Gaza. Discussions on social media revealed widespread confusion, with many citizens expressing uncertainty over which applications were genuinely safe to install. Fear and anxiety were heightened by the timing of the campaign, which coincided with an escalation in missile attacks. The dissemination of false information via platforms such as Facebook and Telegram further disoriented the public, leading some users to disregard legitimate emergency alerts altogether (Saifi & Al-Qadi, 2024).

This phenomenon reflects the erosion of trust that occurs under pressure (Bean & Grevstad, 2023). In times of emergency, avoiding the use of digital tools can increase the risk of physical harm. Particularly when psychological operations are designed to provoke such avoidance, the collapse of confidence in official platforms constitutes a critical vulnerability within national defence infrastructure. The importance of adaptive trust as a cornerstone of cyber resilience has been emphasised (Linkov & Kott, 2019). The HamsaUpdate case posed a widespread and acute challenge to this trust.

4.4 Deficiencies in Institutional Response and Civil Disaster Preparedness

To prevent further spread of the malware, the Israeli National Cyber Directorate (INCD), the Computer Emergency Readiness Team (CERT-IL), and other organisations launched a coordinated response. Indicators of Compromise (IoCs) were provided in detail, and public advisories warned against downloading unfamiliar applications. Malicious links were removed, infected devices were tracked, and filtering algorithms were improved through collaborative efforts between Google and major telecommunications companies. Nevertheless, several critical vulnerabilities were exposed (ENISA, 2023). Firstly, there was a delay in outreach efforts to linguistically diverse populations. Secondly, the general public's level of digital literacy proved insufficient to recognise and counter manipulated content. The absence of prior education regarding social engineering tactics meant that initial reactions were driven more by fear than reason, despite the fact that many citizens eventually removed the malware. As a consequence, the INCD initiated digital hygiene campaigns in the aftermath of the attack, focusing on misinformation awareness and strengthening application verification procedures (INCD, 2024).

This case illustrates that emotional states and levels of social trust significantly influence how civilians engage with emergency digital tools. In particular, public trust in emergency alert systems, and

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the clarity regarding the origin of such messages, determines whether individuals respond appropriately or disregard warnings altogether. A national US survey conducted following the 2021 Wireless Emergency Alert (WEA) test found that, even when citizens did not fully understand alert classifications, they were more likely to trust and act upon WEA messages than other notifications. This underscores the crucial role of message source and content credibility in crisis contexts. These findings align with those of the current study, where confusion surrounding the legitimacy of the HamsaUpdate app led to both hypervigilance and avoidance, ultimately compromising public safety and resilience (Bean & Grevstad, 2022).

Moreover, the findings suggest that even the most advanced technical defences may falter if not supported by comprehensive public education, simulation-based training, and trustworthy real-time communication. In emotionally charged environments, the human factor, particularly trust and comprehension, becomes as pivotal as encryption or malware detection protocols. Strengthening national cybersecurity, therefore, requires not only technological preparedness but also sustained investment in civilian preparedness and psychological resilience.

4.5 More Far-Reaching Implications for Cybersecurity and the Resilience of Alternative Conflicts

In the context of contemporary conflict, the HamsaUpdate incident underscores the complex interplay between malware technology, emotional manipulation, and institutional readiness. As the influence of hybrid warfare intensifies, civilian populations are becoming increasingly central targets. These populations are not merely suffering collateral damage; rather, they are being deliberately subjected to psychological and technological disruption.

The findings of this study support calls to shift national cyber defence doctrine from reactive security to proactive resilience. Priority in policymaking should be given to public training initiatives, inter-agency collaboration, the global exchange of threat intelligence, and the development of culturally adapted digital tools. Effectively addressing future threats requires the integration of technical countermeasures with public education, transparency, and the deployment of digital systems that have been stress-tested under crisis conditions. In addition to contributing to the growing body of literature on civilian cyber risk, this case offers valuable insights for designing robust, citizen-centred cybersecurity frameworks in conflict-affected regions.

5. Discussion

This case study aimed to examine the operational characteristics, psychological impacts, and institutional responses associated with the HamsaUpdate malware campaign, which targeted Israeli civilians during the 2023-2024 Israel-Hamas conflict. The research sought to address a significant gap in the literature by investigating the effects of socially engineered malware campaigns on civilians within hybrid warfare contexts, particularly concerning the interplay between emotional stress, trust in emergency infrastructure, and digital literacy. A document analysis of technical reports, academic publications, and official advisories was employed to elucidate the evolving nature of cyberwarfare that targets both technological systems and societal structures.

As illustrated in Figure 1, the main empirical findings from the HamsaUpdate case align with broader analytical insights, enabling a structured interpretation of how technical mechanisms, psychological responses, and institutional dynamics interact in the context of hybrid cyber conflict.

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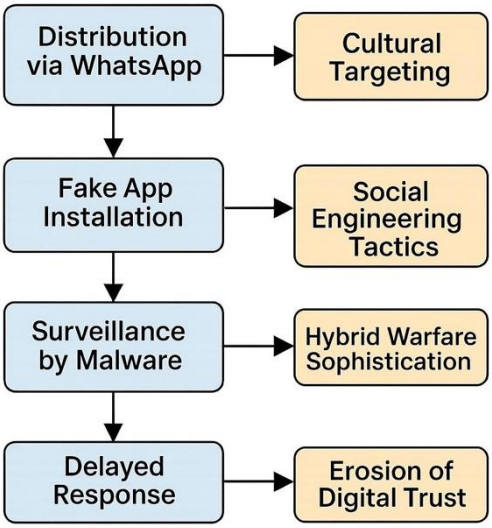
Figure 1. Stages and Impacts of the HamsaUpdate Campaign



5.1 Operational Complexity of the HamsaUpdate Campaign

A principal finding relates to the operational complexity of the HamsaUpdate malware. This campaign exhibited characteristics more commonly associated with state-sponsored cyber-espionage than with conventional financially motivated malware. It employed encrypted command-and-control infrastructure, dynamic IP rotation, and obfuscated code to evade early detection (Check Point Research, 2024; Intezer, 2023). These technical features align with the definition of "military-grade malware", engineered to disrupt civilian life while masquerading as legitimate emergency infrastructure (Kostyuk & Zhukov, 2019).

Moreover, dissemination via widely used platforms such as WhatsApp, coupled with language localisation in Hebrew and Arabic, demonstrated strategic intent grounded in cultural relevance and broad outreach. Culturally tailored social engineering campaigns are more effective at bypassing cognitive defences (ENISA, 2023). This pattern was evident in users mistaking the app for an official Home Front Command product, underscoring the effectiveness of brand imitation and urgency-based psychological triggers. The malware's spread through hijacked accounts and peer-to-peer links further exemplified evasive tactics designed to penetrate the civilian digital ecosystem.

5.2 Emotional Manipulation and Psychological Effects

The psychological dimension of the attack emerged as a dominant theme, surpassing the technical components in terms of societal impact. The campaign deliberately exploited heightened emotional responses during rocket assaults. This manipulation of fear and urgency reflects findings from contemporary psychological cybercrime research, which suggest that emotional dysregulation increases susceptibility to digital deception (Siddiqi et al., 2022; Taddeo & Floridi, 2018).

The HamsaUpdate attack created confusion among civilians, many of whom reported difficulty distinguishing between legitimate emergency alerts and malicious imitations. This phenomenon aligns with the concept of "alert fatigue" described in prior research, a cognitive state in which individuals become desensitised to repeated or contradictory alerts, thereby diminishing their responsiveness to genuine threats (Joseph et al., 2021). During high-stress periods, such as missile attacks or pandemics, the influx of conflicting digital messages can overwhelm users and impair decision-making processes (Shultz et al., 2018). Similar effects of communication overload were observed during the COVID-19 pandemic, where misinformation and inconsistent digital guidance contributed to a decline in public trust in official sources (Brennen et al., 2020). These insights underscore the psychological vulnerabilities that adversaries exploit in hybrid cyber operations.

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5.3 Social Engineering and Cultural Targeting

The attackers employed social engineering in a deliberate and calculated manner. The messages used formal language, incorporated trusted civil defence terminology, and closely resembled official communications from emergency services. This strategy mirrors what has been termed "pretexting with benevolent intent", a tactic whereby attackers pose as trustworthy or protective figures to manipulate victims into compliance (Hadnagy & Schulman, 2021). The HamsaUpdate malware embodied this approach by masquerading as a life-saving emergency application, exploiting individuals' natural inclination to seek safety and authoritative guidance during missile attacks.

The use of indigenous languages, culturally resonant imagery, and contextually relevant emergency scenarios reflects the concept of culturally contextualised cyber deception. This refers to the strategic use of cultural familiarity by attackers to enhance credibility and reduce scepticism (Taddeo & Floridi, 2018). Such techniques are particularly effective in environments characterised by high emotional distress and varying levels of digital literacy. Increasingly, attackers tailor phishing and malware content to reflect the language, visuals, and psychological cues of their target populations, thereby increasing the likelihood of success (ENISA, 2023). This approach enables adversaries to penetrate cognitive defences by embedding deception within familiar cultural and social frameworks, thus bypassing purely technical safeguards.

The HamsaUpdate campaign demonstrates that social engineering is not solely a technical vulnerability, but also a profound cognitive and behavioural one.

5.4 Institutional Reactions and Deficiencies in Public Literacy

The INCD and CERT-IL led the institutional response, which included the dissemination of advisories, IoCs, and public education initiatives. While these measures were proactive, they encountered significant challenges related to timing and inclusivity. Scholars have argued that reactive cyber defence must be supplemented with anticipatory strategies, including public education campaigns and simulation exercises that prepare populations for disinformation attacks (Linkov & Kott, 2019).

This case revealed that delays in reaching Arabic-speaking and socio-economically disadvantaged communities hindered their ability to receive, interpret, and respond effectively to real-time cyber alerts. Such disparities in digital access and literacy extend beyond mere connectivity to include comprehension and contextual relevance, leaving these populations more vulnerable to manipulation during cyberattacks. Research on cybersecurity literacy initiatives has shown that culturally tailored and locally adapted programmes are significantly more effective in raising awareness in under-resourced communities than generic interventions (Ghosh et al., 2023). Without inclusive strategies, such as multilingual training, community-based awareness efforts, and culturally appropriate communication, the architecture of national cyber resilience remains uneven. This inequity enables attackers to exploit both technical and psychological vulnerabilities, placing already marginalised groups at greater risk.

Moreover, the INCD's digital hygiene campaigns were only launched after public trust had already eroded. Had tools for misinformation resilience been implemented earlier and community engagement fostered at the grassroots level, the psychosocial impact of the attack might have been mitigated. This underscores the critical importance of public trust in the effectiveness of national cybersecurity preparedness, particularly in communication and education strategies.

5.5 Civilian Digital Trust and Societal Consequences

This case demonstrates the fragility of digital trust during times of national crisis. For digital emergency systems to operate effectively, the public must have a high degree of confidence in their legitimacy and reliability. Scholars have argued that, when this trust is eroded, civilians may become "behaviourally paralysed", a state in which fear of potential consequences leads to riskier decision-making (Roe et al., 2025).

Malware disguised as assistance constitutes a distinct form of harm that corrodes public trust. This has been conceptualised as "symbolic cyberviolence", an attack that disrupts societal trust in emotional and moral terms (Taddeo & Floridi, 2018). Such cyber harm transcends technical disruption and enters the

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domain of social and ethical injury. It undermines the foundational principles of democratic governance, which depend on secure and trustworthy emergency communication systems.

5.6 Hybrid Warfare as a Civilian-Centered Strategy

The HamsaUpdate campaign exemplifies the strategic shift towards hybrid warfare that deliberately targets civilian populations. Rather than viewing civilians as collateral damage, the campaign placed them at the epicentre of digital conflict. This aligns with the growing thesis that future conflicts will increasingly be "civilian-facing", aiming to influence public morale, perception, and behaviour alongside infrastructure sabotage (Krishnan, 2024). Cyber operations now increasingly seek to undermine societal trust and psychological security, exploiting emotional vulnerabilities to maximise disruption (Taddeo & Floridi, 2018). This evolving form of warfare challenges the conventional divide between military and civilian domains, with civilians becoming direct targets of cognitive and digital manipulation. The case underscores the imperative for cybersecurity strategies to move beyond technical defences and integrate socio-behavioural resilience frameworks.

In conclusion, the HamsaUpdate campaign offers a vivid example of how hybrid warfare materialises through multilayered digital deception. The interplay of social engineering, emotional manipulation, institutional limitations, and civilian vulnerability presents a comprehensive portrait of modern cyber conflict. Building future resilience will require not only advanced technical countermeasures but also the reinforcement of psychological preparedness, digital literacy, and public trust in secure, credible communication systems.

6. Study Limitations

This case study provides critical insights into the operational and psychosocial dynamics of the HamsaUpdate malware campaign; however, several limitations must be acknowledged. The analysis depends on publicly available documents, technical reports, and secondary data sources, which may not fully represent classified operational details or internal institutional dialogues. This reliance may limit the precision of attribution and the technical scope of the malware's full capabilities. The absence of direct access to infected devices or proprietary cybersecurity datasets constrained the study's ability to independently verify forensic findings or conduct advanced reverse engineering of the malware.

The thematic analysis may be subject to inherent interpretative bias associated with qualitative document review. Notwithstanding the utilisation of triangulation across reports from multiple esteemed sources (e.g., Check Point Research, ENISA, INCD), inconsistencies in terminology, focus, and reporting frameworks may compromise thematic coherence. Additionally, the relevance of the findings to other geopolitical contexts is limited. The Israeli context is especially relevant due to its high level of digital literacy and proactive cyber defence approach; however, civilian responses to similar campaigns in less digitally advanced societies may differ.

Finally, the psychological and behavioural implications discussed stem from digital communication and additional behavioural studies. A thorough empirical comprehension requires ethnographic research or survey-based methods involving affected populations. Future research could benefit from access to real-time user data, institutional interviews, and longitudinal observations of trust restoration dynamics after a crisis.

7. Recommendations

Given the HamsaUpdate case and its larger effects, a full plan is needed to strengthen civilian cybersecurity, particularly in the context of hybrid conflicts where civilians are deliberately targeted. First and foremost, it is important to create long-lasting national programmes that will help people become more digitally literate and address the psychological aspects of cyber manipulation. These programmes need to be specifically designed for times of national crisis, when people are more likely to be emotionally manipulated and mentally overloaded. As part of public education programmes, people should learn how to recognise socially engineered communications, distinguish between legitimate and fake apps, and stay

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calm during emergencies. To ensure inclusive digital readiness, it is important to intentionally interact with communities that speak different languages and have different cultures.

Second, there needs to be improved formal coordination between government agencies, private cybersecurity companies, telecommunications companies, and the major app distribution platforms. The timely sharing of threat intelligence, along with clear protocols for app removal and public notification, can significantly accelerate the response to cyber threats. Cybersecurity incident response units need to be able to address both technical and behavioural dimensions of attacks. This includes the capacity to monitor public reactions and manage confusion or distress during cyber incidents. Setting up multi-agency task forces with clear roles for dealing with cyber crises will make responses faster and more consistent.

Zero-trust security models and multi-factor authentication protocols must be required for all important infrastructure sectors and public service platforms. These steps are necessary to reducing the effectiveness of malware attacks, especially on mobile devices. It is also important to use principles from behavioural science when designing user interfaces. Alerts, permission requests, and update prompts must be designed to discourage impulsive responses, especially under stress. This can be achieved by using visual and verbal cues that slow down decision-making when things are stressful. This can help users question, rather than give in to, bad prompts.

Additionally, simulation-based training exercises should be conducted regularly, not only for cybersecurity professionals but for the general public. These exercises should simulate real-life mobile malware and phishing situations, allowing individuals to practise identifying threats and responding to them in controlled settings. Schools, community groups, and local governments can all work together on this public resilience project. These simulations should not only educate but also reduce anxiety, build digital confidence, and foster a sense of control during online uncertainty.

To ensure that cyber alerts are disseminated quickly, clearly, and reliably, national crisis communication protocols need to be improved. During the HamsaUpdate campaign, people were confused because communication was delayed and inconsistent. To address this, alert systems need to be easy to understand in terms of language, visually differ from regular apps, and have official institutional branding. Adding these design features to civil defence apps can make people more likely to trust them and reduce susceptibility to spoofing attempts.

The recommendations outlined above call for a shift from cyber defence strategies that are reactive and focused on technology to a resilience framework that is proactive and focused on people. This includes mental preparedness, improving collaboration between institutions, using design to keep people safe, and making cyber awareness a part of everyday life. Only through an integrated national strategy can societies protect both their technological infrastructure and the public trust that hybrid warfare seeks to erode.

8. Conclusions

The HamsaUpdate malware campaign is a good example of how hybrid warfare is changing in the digital age. Unlike most cyberattacks that primarily target infrastructure or pursue financial gain, this campaign was deliberately designed to manipulate emotions, exploit psychological vulnerabilities, and erode public trust in official digital systems during a time of national crisis. The operation showed that technology can be used to attack civilians through a fake civil defence app during an active conflict, which weakens the social fabric that supports societal resilience.

The results of this study show that socially engineered malware represents a complex and multifaceted threat. The danger of such attacks lies not only in their technical sophistication but also in their ability to use cultural signals, exploit fear, and pose as legitimate safety tools. The documented psychological and behavioural effects, especially confusion, fear, and the loss of trust in technology, underscore the importance of expanding the understanding of cybersecurity to include emotional and cognitive factors.

Although some institutions responded quickly, the responses revealed a lack of public preparedness, especially among linguistically diverse and digitally underserved populations. These challenges highlight that technical measures alone are insufficient for ensuring cybersecurity. Without public trust, transparent

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communication, and proactive education, even the most advanced technologies may fail. The reactive nature of the initial response emphasises the need to build robust infrastructure that can both adapt to and anticipate evolving tactics in contemporary cyber campaigns.

Furthermore, the case showed that the main goals of the conflict had changed. In digital warfare, civilians are no longer just watching; they are now a key part of the strategy. The deliberate focus on how civilians act, what they think, and how they build trust illustrates a change in the way people think about modern conflicts. The battlefield is now both digital and emotional, and the effects go beyond physical to include mental and social aspects.

This study suggests that national security strategies need to adapt to include methods that focus on the public, such as technological defences, societal cohesion, behavioural readiness, and cultural awareness. The combination of malware, false information, and social unrest creates a dangerous situation that calls for interdisciplinary approaches that combine cybersecurity skills with behavioural science, communication strategies, and public policy.

The HamsaUpdate incident is both a warning and a lesson. It highlights the vulnerability of technologically advanced societies to hybrid threats that make people lose faith in the government. It urges organisations to reconsider not only how to detect and respond to such attacks, but also how to foster more resilient social systems that are less susceptible to psychological manipulation. In a time when wars may start on a screen, it becomes clear that building digital resilience is not a luxury but a basic part of national defence.

Declarations

Competing interests' declaration: No potential conflicts of interest identified.

Data availability statement: Data supporting the study can be accessed on request from the authors.

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